Course Description

Prof. Dr. Gerardo Flores

January 2026

Microprocessor Systems (CSCE 3340 - SENG 3345)

TAMIU

Microprocessor Systems (CSCE 3340 - SENG 3345)

Times and Location

MWF 1:10pm-2:05pm in Billy F Cowart Hall 203

Course Description

Basic computer structure, the instruction set, addressing modes, assembly language programming, assembly language subroutines, arithmetic operations, programming in C, implementation of C procedures, elementary data structures, input and output and a survey of microprocessor based design. This course is interchangeable with SENG 3345.

Prerequisite: CSCE 2330 Engineering Department, College of Arts & Sciences

Additional Course Information:

- Introduction to microprocessor systems with **assembly language** and **C programming**.
- Focus on **microprocessor architecture**, peripheral programming, and **hands-on practice**.
- Use of simulation tools: **Cpulator**, **Keil µVision**, **STM32CubeIDE**, and **Wokwi**.
- Quick review of basic concepts to work with **ARM Cortex microcontrollers**.
- AI tools encouraged for learning but not permitted during exams.

Program Learning Outcomes

Foundational Concepts:
•Understand differences between microprocessors and microcontrollers.
•Explore Harvard vs. Von Neumann architectures.
•Proficiency in data representation (binary, hexadecimal, signed/unsigned integers).

Assembly and C Programming:
•Write, debug, and integrate assembly and C programs for ARM Cortex-M microcontrollers.
•Practice arithmetic, logical operations, control flow, and memory access.

Simulation and Tools:
•Use Cpulator, Keil µVision and Wokwi to simulate and test functionality like **GPIO**, **ADC**, **PWM**, and **interrupts**.

Hardware-Software Integration:
•Configure and program peripherals (GPIO, timers, ADC, PWM).
•Manage interrupts (NVIC, ISRs).

Embedded System Design:
•Design, simulate, and implement embedded systems.
•Apply debugging and problem-solving techniques.
•Complete a project combining theoretical knowledge with real-world applications.

Student Learning Outcomes

1. Develop a solid understanding of microprocessor architecture and programming, including assembly language and C for **ARM Cortex-M microcontrollers**.

2. Gain proficiency in using industry-standard tools like **Cpulator**and **Wokwi** to simulate, program, and debug embedded systems.

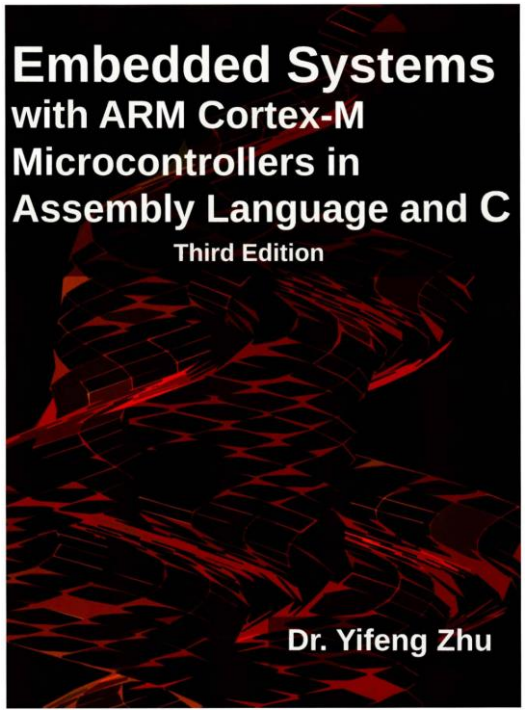
3. Design and analyze embedded systems for various applications, focusing on peripheral integration, interrupts, and hardware-software interaction.

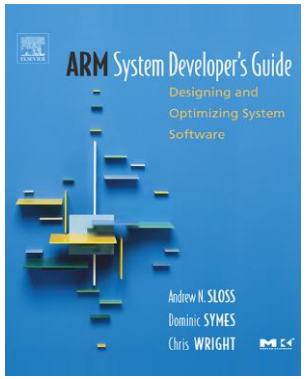
Important Dates

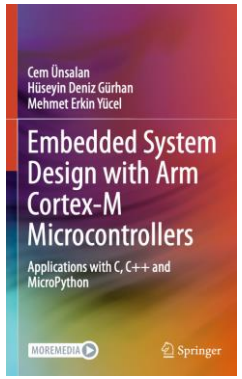
Visit the Academic Calendar (tamiu.edu) (https://www.tamiu.edu/academiccalendar/) page to view the term's important dates.

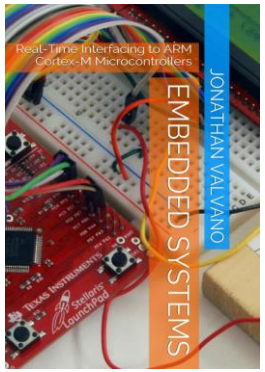
Textbooks

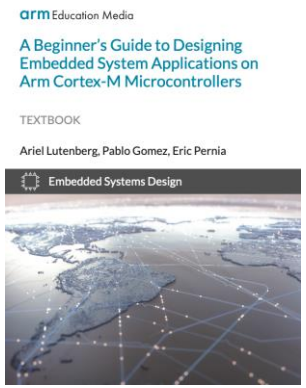
Group	Title	Author	ISBN
Optional	Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C (Third Edition)	Yifeng Zhu	9780982692660
Optional	The Definitive Guide to the ARM Cortex-M3 (Second Edition)	Joseph Yiu	978-1-85617-963-8
Optional	ARM System Developer's Guide: Designing and Optimizing System Software	Andrew N. Sloss, Dominic Symes, Chris Wright	978-0121477513
Optional	Embedded System Design with ARM Cortex-M Microcontrollers	Cem Ünsalan, Hüseyin Deniz Gürhan, Mehmet Erkin Yücel	978-3-030-88438-3
Optional	A Beginner's Guide to Designing Embedded System Applications on ARM Cortex-M Microcontrollers	Ariel Lutenberg, Pablo Gomez, Eric Pernia	9781911531418
Optional	Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers	Jonathan Valvano	978-1463590154
Optional	Mastering STM32	Carmine Noviello	NA

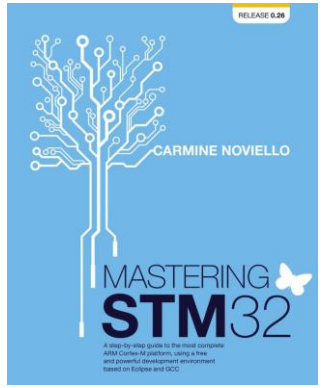


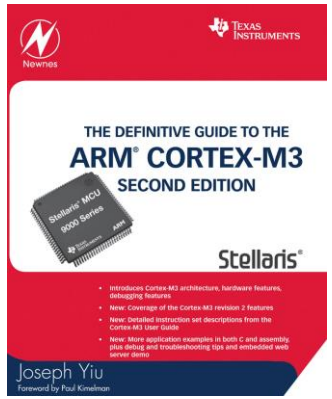












Other Course Materials

1. Cpulator Simulator: For developing and debugging assembly and C programs for microcontrollers.

2. Wokwi Simulator: An online platform for simulating embedded systems, allowing students to program in C and visualize peripherals such as GPIO, ADC, and PWM.

3. STM32CubeIDE (optional): An integrated development environment for STM32 microcontrollers, supporting both C programming and debugging features.

Additionally, students will have access to notes, reference materials, and documentation for ARM Cortex-M microcontrollers. These resources will provide both theoretical and practical insights into microprocessor architecture, programming, and real-world applications.

Students are encouraged to install and familiarize themselves with these tools before the start of the course to maximize hands-on learning.

Grading Criteria

- Final Exam – 25%: A comprehensive final exam that assesses the overall knowledge of the course.

- Midterm (2 Key Assignments or Exam) – 30%: Two key assignments or a midterm exam that together account for 30% of the total grade, with each
- component (assignment or exam) representing 15% of the overall grade.

- Project, Participation, Short Assignments, and Attendance – 45%: Students will collaborate in teams of 5 to n members to develop a comprehensive project that spans the entire semester. Also, this component reflects active class participation, completion of short homework assignments, and consistent attendance.

This project is an opportunity to apply the knowledge and skills gained throughout the course in a meaningful and practical way.

Teams have the flexibility to choose from a range of project types, including:

1. **Project Work: Microcontroller-Based Specialized Task**
   - Propose and implement a microcontroller project to address real-world problems.
   - Integrate features like GPIO, ADC, PWM, or interrupts using C or assembly programming.
   - Deliverables: functioning prototype, brief report, and presentation showcasing design, challenges, and improvements.
2. **Presentation**
   - Research and deliver a professional presentation on a specialized area of ARM Cortex microcontrollers.
   - Base content on selected chapters from the course textbooks, with simulations when needed.
3. **Simulation Project: Specialized Task Using Wokwi**
   - Design and simulate a microcontroller-based system using the Wokwi platform.
   - Incorporate features like GPIO, ADC, PWM, or interrupts.
   - Deliverables: complete simulation setup, concise report, and presentation of functionality, challenges, and potential extensions.
4. **Goals**
   - Encourage collaboration, innovation, and practical application.
   - Explore areas of personal interest in microprocessors, preparing students for real-world applications and professional development.

Grading Criteria

GRADE	PERCENTAGE
A	91-100
B	80-90.9
C	70-79.9
D	60-69.9
F	Below 60

Assignments

- Tasks include readings, research, exercises, and coding challenges aligned with lecture topics.
- Designed to reinforce understanding and provide hands-on experience with theoretical concepts.
- Some assignments may offer **extra credit**, evaluated on complexity and effort, to boost final grades.

Schedule of Topics and Assignments

Week of	Agenda/Topic	Reading(s)
1/21	Introduction to the Course	• Course overview and objectives
1/26	Basic Concepts	• Microprocessors vs. Microcontrollers • Harvard and Von Neumann architectures
2/2	Data Representation	• Bits, bytes, words, and double words • Binary, octal, decimal, and hexadecimal number systems • Unsigned and signed integers: Sign-and-magnitude. One's complement and two's complement
2/9	Introduction to Assembly Programming	• ARM instruction set overview: Instruction format. The basic structure of an assembly program • Assembly directives
2/16	Arithmetic and Logical Operations	• Basic operations: Addition, subtraction, multiplication, and division
2/23		• Logical operations: AND, OR, NOT , XOR • Bit manipulation: Shifting and rotating
3/2	Midterm	
3/9	Spring Break	No classes
3/16	Memory Access and Register Operations	• Load and store instructions: LDR, STR, MOV • Addressing modes: Pre-index, post-index, and PC-relative addressing
3/23	Control Flow	• Branch and conditional execution: Branch, if-then, for, while loops
3/30	Subroutines and Stack Management	• Calling subroutines • Stack operations: PUSH and POP • Preserving the runtime environment • Calling C functions from assembly and vice versa
4/6	Interrupts and Peripheral Control	• Basics of interrupts • NVIC configuration in ARM Cortex-M
4/13	Digital Input and Output	• GPIO configuration and usage
4/20	Analog-to-Digital Conversion (ADC)	• Fundamentals of ADC • Configuring and reading from an analog sensor
4/27	Final projects week	• Final projects and exercises for preparation of final exams
5/4	Reading Day	No classes
5/11	Final Examination period	